

Percentile and Quartiles

⇒ Percentage of {1, 2, 3, 4, 5, 6}

No. of odd numbers = 3.

Percentage of odd no. in this group = $\frac{3}{6} \times 100 = 50\%$

⇒ Percentile: It is a value below which a certain percentage of observations lie.

{2, 2, 3, 4, 5, 5, 6, 7, 8, 8, 9, 9, 10}

$n = 12$

Percentile of value, $x = \frac{\text{No. of values below } x}{n} \times 100.$

Percentile ranking $\leftarrow = \frac{11}{14} \times 100 = 78.57\%$ of value 9

$$\Rightarrow \text{Value} = \frac{\text{Percentile} \times (n+1)}{100} = \frac{3+4}{2} = 7$$

$$\text{Value} = \frac{25 \times (14+1)}{100} = 3.5$$

$$= \frac{25 \times 15}{100} = 3.75 \approx 3.5$$

$\Rightarrow$ Quartile :-

(15) P, 8, 8, 5, 2, 2, 2, 2, 2, 2, 2, 4, 5, 8

25% = 1st Quartile

50% = 2nd Quartile

75% = 3rd Quartile

